MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

CSIT DEPARTMENT

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Course name:	Data Mining (DATM-6200)
Title	Lab 7
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1. **A short description of the two datasets.**
   The file ‘malawi_retail_monthly_timeseries.csv’ has 60 rows and 5 columns, each row represents a month from January 2021 to December 2025. The dataset records four economic indicators for a Malawian retail context: monthly retail sales in Malawian Kwacha (MWK), the average pump price of fuel (MWK/litre), the national inflation rate (%), and a tourism activity index. Here is a table depicting the dataatypes and attributes

Columns	**Datatype**
month	date
avg_fuel_price_mwk	float
inflation_rate_pct	float
tourism_index	float
monthly_sales_mwk	float

Another separate file was provided aswel, the file called ‘store_event_ stream_sample.csv’. This file has 5,000 rows and 5 columns, and each row represents a single monitoring event recorded at one of 12 retail stores between 1 June 2025 and 18 June 2025. Each event captures the store identifier, the instantaneous queue length, the number of transactions processed in the last 15 minutes, and any stock-level alerts raised.

Columns	**Datatype**
event_time	datetime
store_id	string
queue_length	int
transactions_last_15min	int
stock_alerts	int

2. The preprocessing steps used for date/time handling.
   Initially, as per given files, the ‘month’ column was recorded in ISO string format (Year-Month-Day). Inorder to process this, we had to parse it into pandas datetime object using pd.to_datetime() function, then sorted it ascending by month.

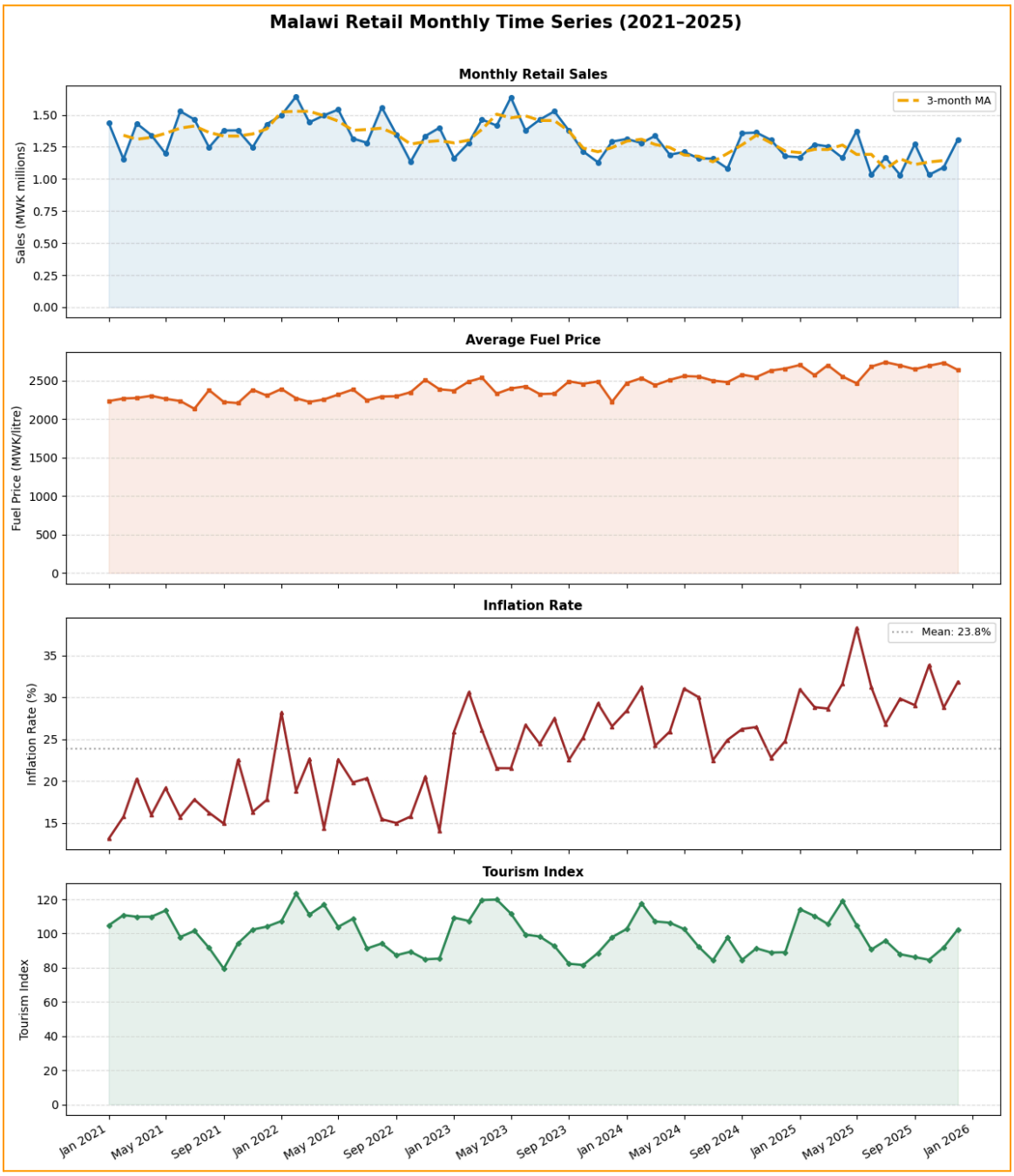
   Then the column called ‘event_time’ was parsed from string to datetime and them events sorted

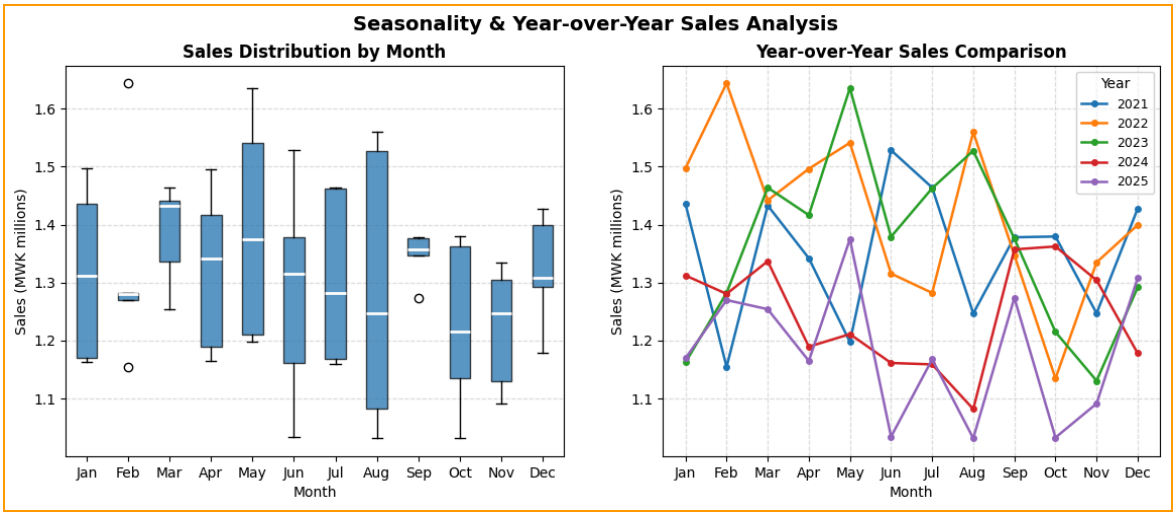
   The drop = true attribute inside reset index is just for resetting the index

Preparing the time variables

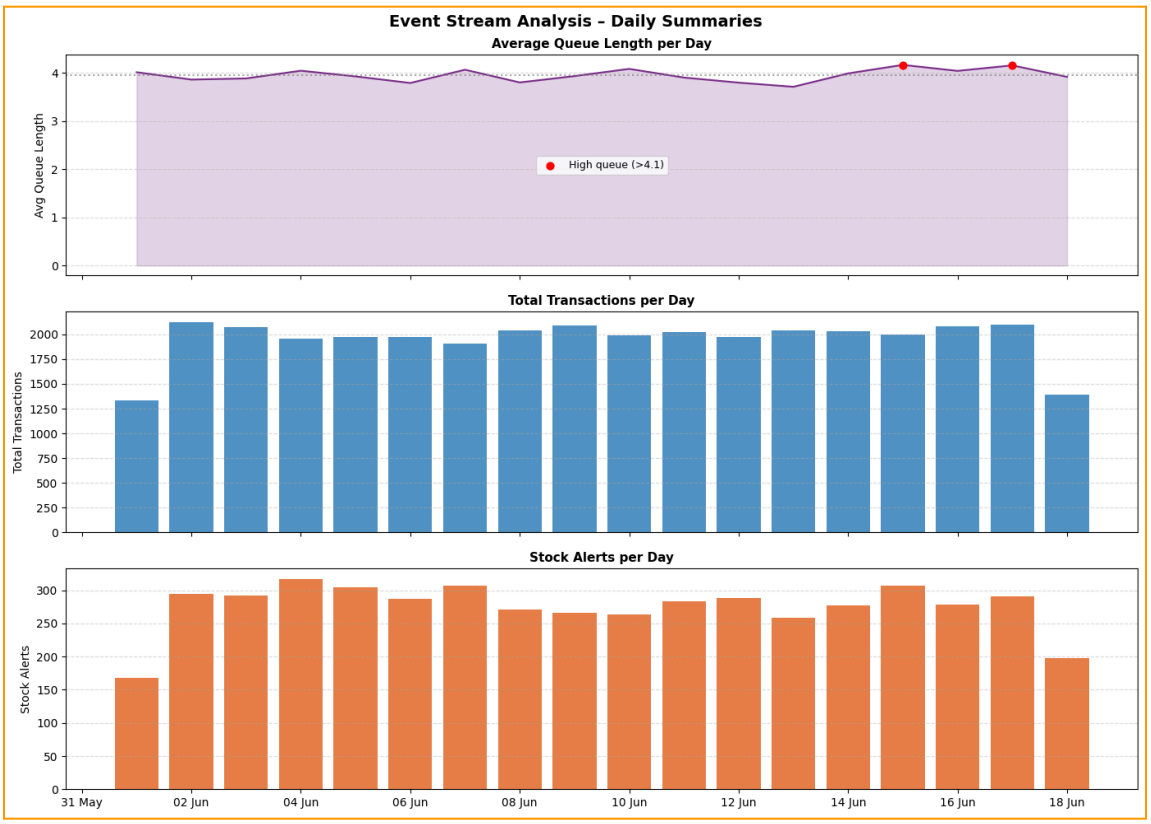
```
ts['month'] = pd.to_datetime(ts['month'])
ts = ts.sort_values('month').reset_index(drop=True)

ev['event_time'] = pd.to_datetime(ev['event_time']).dt.floor('h')
ev = ev.sort_values('event_time').reset_index(drop=True)
```

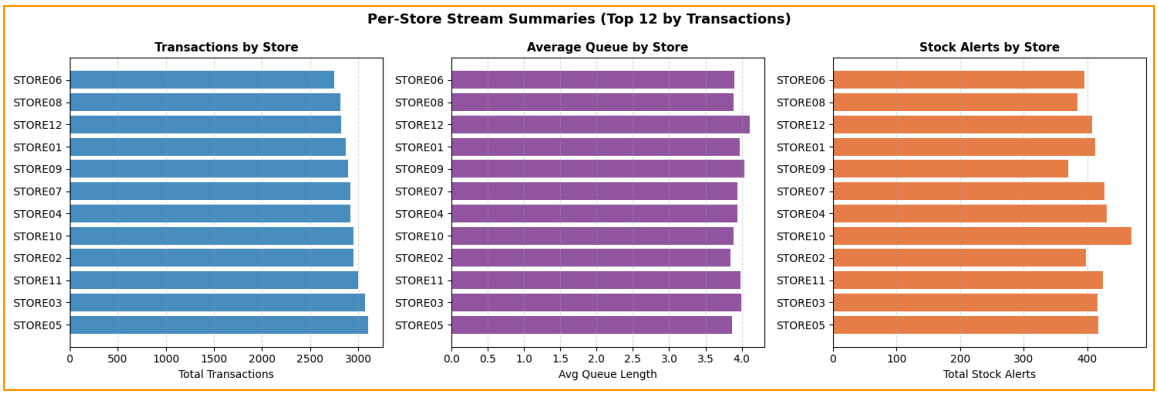
3. At least three visualizations from the monthly time series.



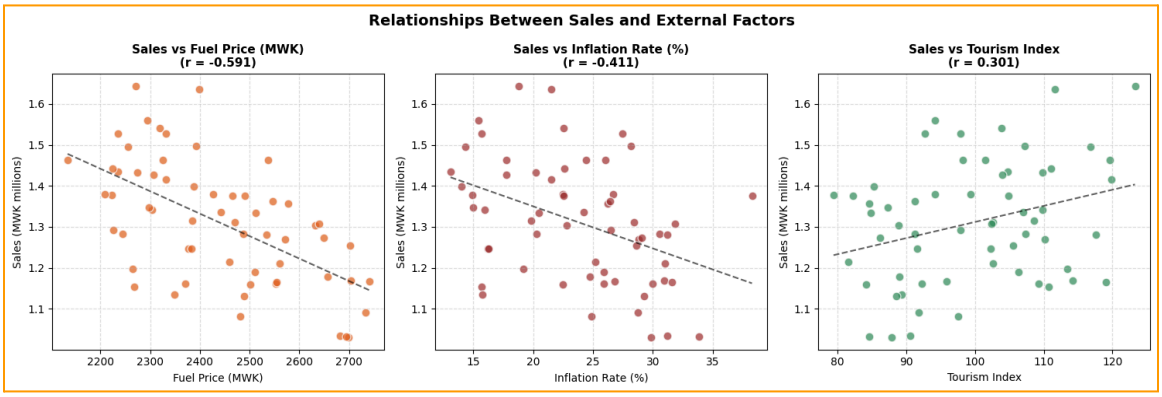


4. At least two summaries or visualizations from the event stream data.





5. A short explanation of any trend, seasonality, or unusual events observed.




```
=== CORRELATIONS WITH SALES ===
  Sales vs avg_fuel_price_mwk: r=-0.5907
  Sales vs inflation_rate_pct: r=-0.4114
  Sales vs tourism_index: r=0.3010

=== STREAM STATS ===
  Total events: 5000
  Date range: 2025-06-01 08:00:00 to 2025-06-18 16:35:00
  Stores: 12
  Max queue: 13 | Mean: 3.95
  Total stock alerts: 4952
  Hourly peak tx: 2123 at 2025-06-02 00:00:00
  High-queue events (>mean+1.5sd): 2
```

Using the above charts, there is negative correlation between fuel price and sales, in other words, when fuel price is increasing, sales decrease, as as when inflation rate increases, sales decrease too. There is a a positive correlation between tourism index and sales, in other words, the more tourists are attracted, the higher the sales

6. A short interpretation (8–12 lines) of the practical meaning of the results
   Aside from the interpretations which has been given in Q5, which is the above question, here are more observations;

- There were bursts that clustered around mid-morning (10:00–12:00) and mid-afternoon (14:00–16:00), aligning with customer lunch-break and after-work shopping windows.
- Transaction Spikes: The single busiest hour in the dataset (2 June 2025, 11:00) logged 116 transactions,